

Mingzhi Chen

[Email](#)[GitHub](#)[ORCID](#)[Personal Website](#)

Education

Hangzhou Dianzi University (HDU), Hangzhou, China

Sep. 2023 - Jun. 2027 (Expected)

B.S. in Artificial Intelligence | GPA: **4.43/5.00** | Average: **89.6/100** | Top 10%

Selected Coursework: Pattern Recognition & ML (96), Signals & Systems (97), Computer Vision (90)

Publications and Manuscripts

- Mingzhi Chen**, Zhaoyang Xu, Gaopeng Huang, Guangyuan Zhang, Wenwen Tang, Tao Zhang, Zhao Huang, Wenbin Zhang, Huiyu Zhou, Jin Liu, Zhiwen Zheng, Shaowei Jiang, Xiaoshuai Zhang, and Xingru Huang. “**PinNet: Edge-Optimized Skin Lesion Segmentation.**” *Expert Systems with Applications*, 2026. doi:10.1016/j.eswa.2026.134354.
- Mingzhi Chen**, Liqi Yan, Fangli Guan, Bin Lin, Dingran Dong, and Pan Li. “**ToGo: Planning Dexterous Tool Use via Tool-Centric Functional Goal-Trajectory Generation.**” Under review at the *AAAI Conference on Artificial Intelligence (AAAI 2027)*, submitted Jul. 2026.

Research Experience

Tool-Centric Planning for Dexterous Tool Use

Mar. 2026 - Jul. 2026

- Formulated dexterous tool use as tool-centric functional goal-trajectory generation, representing manipulation tasks through functional tool geometry and tool-environment interaction constraints.
- Developed a VLM-grounded geometric planning pipeline that grounds functional tool parts and interaction targets from language, tool geometry, and RGB-D observations, then compiles them into staged SE(3) tool-pose goals for execution by a pretrained goal-tracking policy.

Non-Line-of-Sight Reconstruction via Target-Domain Latent Alignment

Aug. 2025 - Nov. 2025

- Formulated severely degraded NLOS reconstruction as a target-domain latent alignment problem, using representations learned from clean target images as a structural prior for recovering information obscured by diffuse light transport.
- Decoupled target representation learning from measurement inversion by first learning a compact target latent space and its decoder, then training the NLOS encoder to align measurements with this latent space through coupled latent- and image-space objectives.

PinNet: Edge-Optimized Skin Lesion Segmentation

Jul. 2024 - Dec. 2024

- Designed an ultra-lightweight segmentation architecture around the boundary ambiguity and multi-scale appearance variation of skin lesions, combining morpho-spectral feature extraction with dual-perspective attention to strengthen edge-sensitive and contextual representations.
- Reduced the model to approximately 0.05M parameters and 0.068 GFLOPs while maintaining competitive segmentation performance, targeting deployment under strict computation and memory constraints.

National Undergraduate Innovation Training Project

Mar. 2024 - Dec. 2024

- Successfully proposed and secured RMB 10,000 in national-level undergraduate research funding as project lead for an early-stage study on lightweight medical image segmentation.
- Led the project from problem formulation and model development to experimental evaluation, technical documentation, and final delivery, establishing the foundation for subsequent research on efficient skin lesion segmentation.

Honors and Awards

National Third Prize, National Robotics Creative Competition

2025

Third-Class Scholarship (Top 10%), Hangzhou Dianzi University

2023-2025

National Undergraduate Innovation Training Project (National Level)

2024

Granted Software Copyright (1)

2024

Skills

Programming Languages: Python, C/C++, MATLAB

Machine Learning: PyTorch, NumPy, OpenCV

Robotics Tools: Isaac Gym, MuJoCo, LeRobot

Robot Platforms: SO-101

Developer Tools: Git, Linux, LaTeX